

Salesforce Project Documentation

RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions

Implementation Report Based on Completed Work

Project Information	Details
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Project	RePlastix Innovations
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Scope of this report: the Salesforce components configured, automated, tested, and documented as part of the RePlastix Innovations project.

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PHASE 1 — REQUIREMENT ANALYSIS & PLANNING

1.1 Business Objectives

RePlastix Innovations is designed to provide a centralized Salesforce solution for managing the operational lifecycle of plastic waste and recycled products. The primary objective is to organize collection records, recycling-center information, product inventory, customer orders, and restock requests in one connected system.

The project also aims to reduce manual follow-up by introducing automated low-stock monitoring and programmatic inventory processing. The system is intended to improve visibility of stock levels, prevent invalid transactions, and support timely replenishment when customer demand exceeds available product stock.

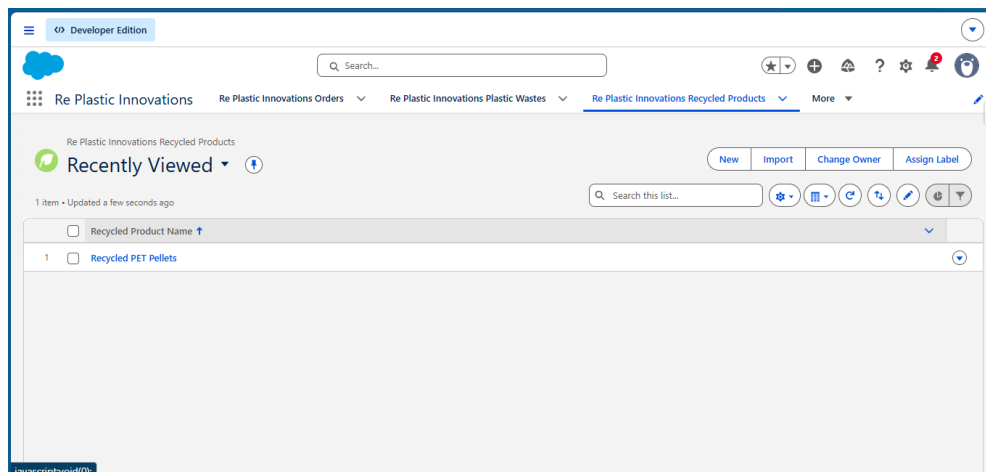
1.2 Functional Scope

The implemented scope covers the major records and processes required for the project. The solution uses five custom objects and their relationships to represent the business process.

- Plastic Waste — records collected plastic waste, including weight, type, collection date, status, location, and recycling center.
- Recycling Center — stores recycling-center details, location, and processing capacity.
- Recycled Product — maintains product stock, threshold, price, and calculated stock status.
- Order — records customer orders, selected recycled product, quantity, and delivery information.
- Restock Request — records shortages, requested quantity, and restock status.
- Automation covers daily low-stock task creation, order-based stock processing, restock-request creation, approved restock processing, and email notification.

Figure 1: RePlastix Innovations Salesforce Application and Main Navigation

Fig. 1a – Recycled Products list



1.3 Stakeholder Mapping

The implementation separates responsibilities through Salesforce roles and profiles. The hierarchy and object permissions are designed so that operational users receive the access required for their work while sensitive records remain restricted by default.

Stakeholder / Role	Primary Responsibility	Relevant Access
CEO	Overall oversight	Higher-level record visibility
Recycling Manager	Plastic-waste operations	Plastic Waste create/read; shared waste records
Sales Representative	Customer orders and products	Order and Account create/read; product read
Warehouse Supervisor	Inventory and restocking	Custom-object create/read/edit; restock visibility

Figure 2: Role Hierarchy and User/Profile Configuration

Fig. 2a – Role hierarchy

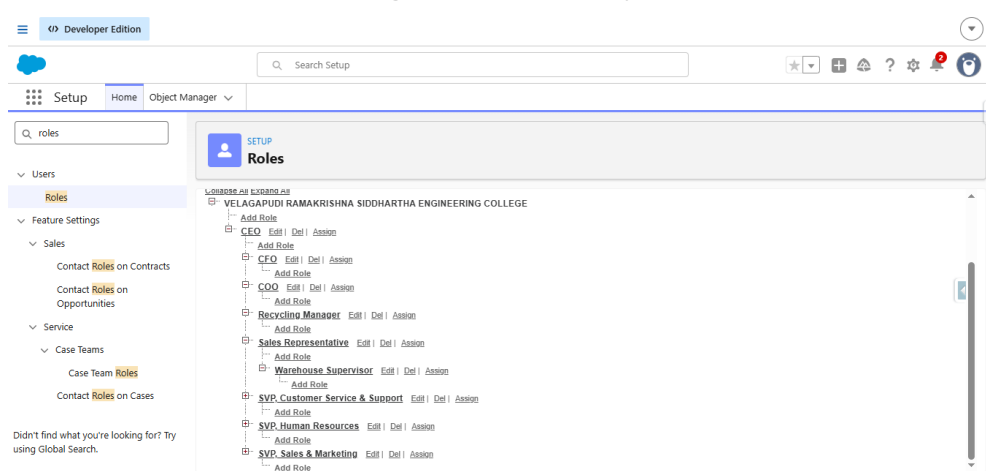
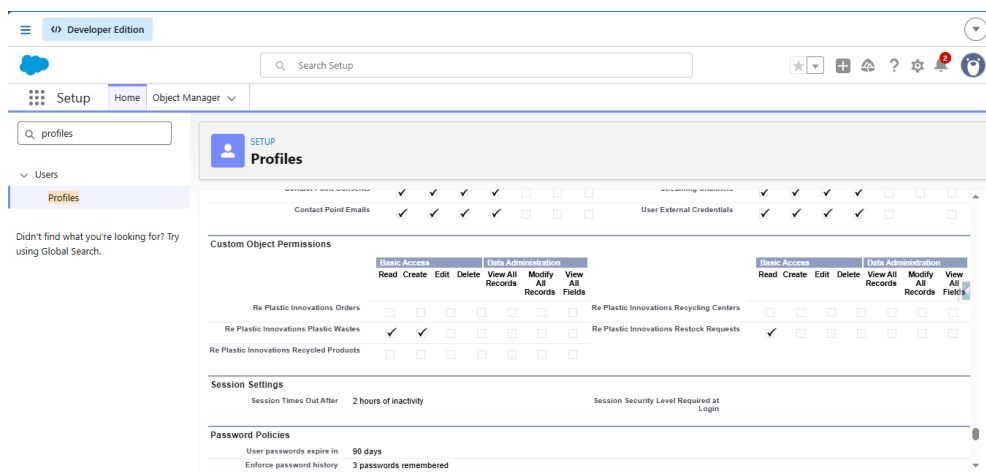


Fig. 2b – Profiles and users



1.4 Execution Roadmap

The implementation was completed in a sequence that moved from data design to application configuration, automation, programmatic processing, security, and testing. This order reduced dependency problems because relationships and fields were established before automation and Apex logic were tested.

- Define business requirements and identify core Salesforce objects.
- Create custom objects, fields, picklists, relationships, and tabs.
- Configure the Lightning application and navigation.
- Implement formula and validation rules for business data quality.
- Create scheduled Flow automation for low-stock monitoring.
- Implement Apex inventory processing and triggers for orders and restock approvals.
- Configure profiles, roles, private record access, and sharing rules.
- Create and execute the required test scenarios and verify successful results.

PHASE 2 — SALESFORCE DEVELOPMENT – BACKEND & CONFIGURATIONS

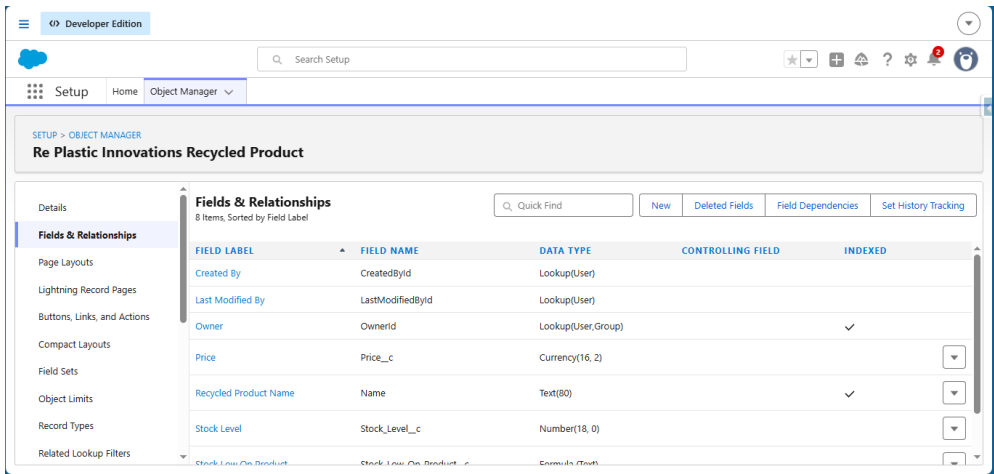
2.1 Data Architecture

The data model uses separate custom objects for the main business entities. Lookup relationships connect records without combining unrelated responsibilities into one object. This structure supports easier maintenance and allows users to navigate from a product to related orders or restock requests.

Object	Important Fields	Relationship
Plastic Waste	Weight, Type, Collection Date, Status, Location	Lookup to Recycling Center
Recycling Center	Location, Capacity	Referenced by Plastic Waste
Recycled Product	Stock Level, Threshold, Price	Referenced by Order and Restock Request
Order	Customer, Product, Quantity, Delivery Date	Lookup to Account and Recycled Product
Restock Request	Product, Requested Quantity, Status	Lookup to Recycled Product

Figure 3: Custom Object and Field Configuration

Fig. 3a – Recycled Product fields and relationships



2.2 Automation Logic

The project uses declarative and programmatic automation according to the type of requirement. A scheduled Flow handles the time-based low-stock check, while Apex triggers handle transaction-based inventory processing that requires related-record queries, calculations, and updates.

Automation	When it runs	Main action
Scheduled Flow — Stock Level Is Low	Daily at 6:00 AM	Checks Stock Level against Threshold and creates a High-priority Task when stock is low
UpdateStockAfterOrder	After Order insert	Calls InventoryManager to reduce stock or create a shortage restock request
UpdateStockAfterRestockApproval	After Restock Request update	Detects Pending → Approved, increases stock, and calls email notification helper

Figure 4: Scheduled Flow – Stock Level Is Low

Fig. 4a – Scheduled Flow – Stock Level Is Low

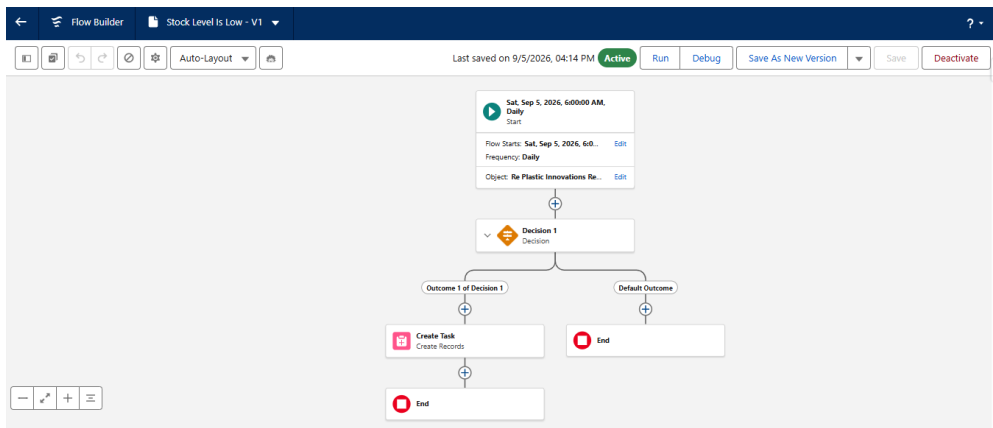


Figure 5: Order and Restock Trigger Workflow

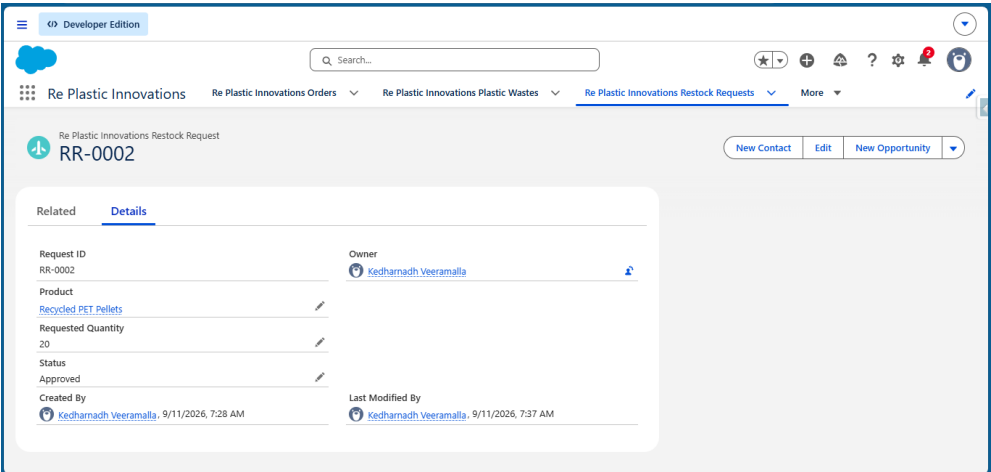
Fig. 5a – Customer order record

The screenshot shows the 'Customer order record' for 'Re Plastic Innovations Order ORD-0004'. The interface includes a top navigation bar with 'Developer Edition' and a search bar. The main content area displays the order details under the 'Details' tab. The order ID is 'ORD-0004'. The owner is 'Kedharnadh Veeramalla'. The customer is 'Veeramalla, Kedharnadh'. The recycled product is 'Recycled PET Pellets'. The quantity is '150'. The delivery date is not specified. The order was created by 'Kedharnadh Veeramalla' on '9/11/2026, 7:28 AM' and last modified by 'Kedharnadh Veeramalla' on '9/11/2026, 7:28 AM'. The interface also includes a 'New Contact' button and a 'Ne' button.

Fig. 5b – Restock request – Pending

The screenshot shows the 'Restock request – Pending' for 'Re Plastic Innovations Restock Request RR-0002'. The interface includes a top navigation bar with 'Developer Edition' and a search bar. The main content area displays the request details under the 'Details' tab. The request ID is 'RR-0002'. The owner is 'Kedharnadh Veeramalla'. The product is 'Recycled PET Pellets'. The requested quantity is '20'. The status is 'Pending'. The request was created by 'Kedharnadh Veeramalla' on '9/11/2026, 7:28 AM' and last modified by 'Kedharnadh Veeramalla' on '9/11/2026, 7:28 AM'. The interface also includes a 'New Contact' button, an 'Edit' button, and a 'New Opportunity' button.

Fig. 5c – Restock request – Approved



2.3 Business Rules

Business rules are implemented through formulas, validation rules, Flow decisions, and Apex logic. The design keeps simple field-level rules declarative and uses Apex for multi-record transaction processing.

- Stock status formula displays 'Low Stock - Restock Needed' when Stock Level is below Threshold; otherwise it displays 'Sufficient Stock'.
- Order quantity must be greater than zero.
- Plastic Waste collection date cannot be in the future.
- When sufficient stock exists, an order reduces the related product stock by the ordered quantity.
- When stock is insufficient, the system creates a Pending Restock Request for the shortage amount.
- When a restock request changes to Approved, the product stock is increased by the requested quantity and the notification helper is invoked.

Figure 6: Formula Field and Validation-Rule Configuration

Fig. 6a – Stock-low formula field

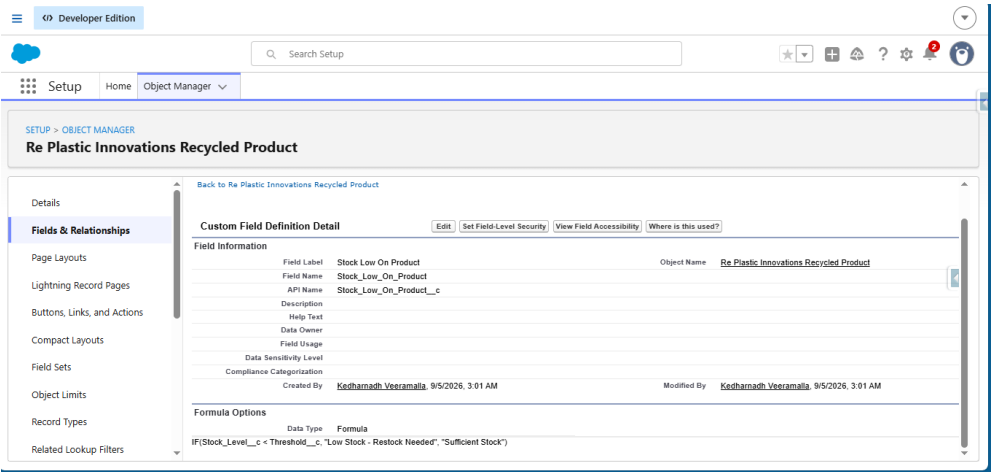


Fig. 6b – Order quantity validation rule

PHASE 3 — UI/UX DEVELOPMENT & CUSTOMIZATION

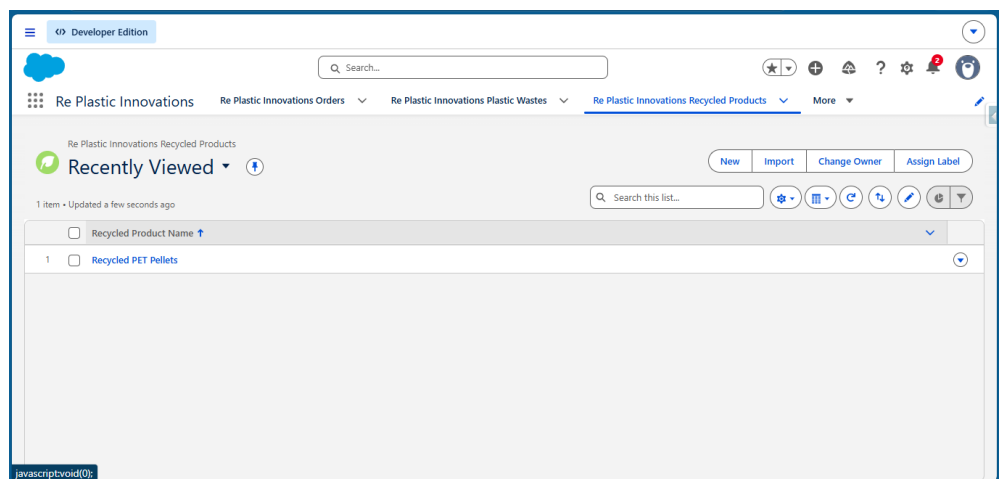
3.1 Interface Design

A dedicated Lightning application named Re Plastic Innovations was created as the central workspace. Custom tabs were created for the project objects so that the main business records are directly accessible from the application navigation.

The interface is organized around the operational workflow rather than around technical configuration. Users can move between waste records, recycling centers, recycled products, orders, and restock requests without navigating through Salesforce Setup.

Figure 7: RePlastix Innovations Lightning App

Fig. 7a – RePlastix Lightning application



3.2 Navigation Flow

The navigation flow follows the business process. Waste records can be associated with recycling centers, recycled products hold the resulting inventory, customer orders consume available stock, and restock requests handle shortages.

Starting Point	Next Record / Action	Purpose
Plastic Waste	Recycling Center	Associate collected waste with a processing location
Recycled Product	Order	Process customer demand against product stock
Order	Restock Request	Handle a shortage when available stock is insufficient
Restock Request	Recycled Product	Increase stock after the request is approved

Figure 8: Application Records and Relationships

Fig. 8a – Plastic Waste record

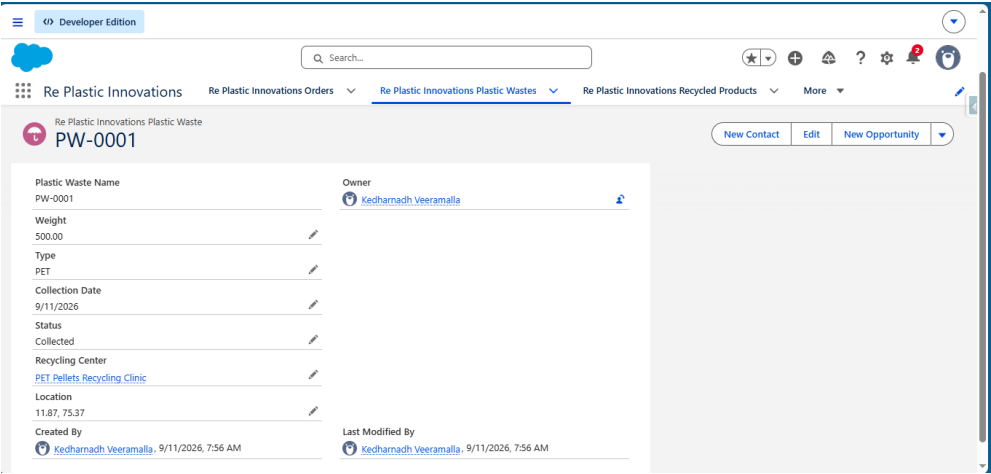
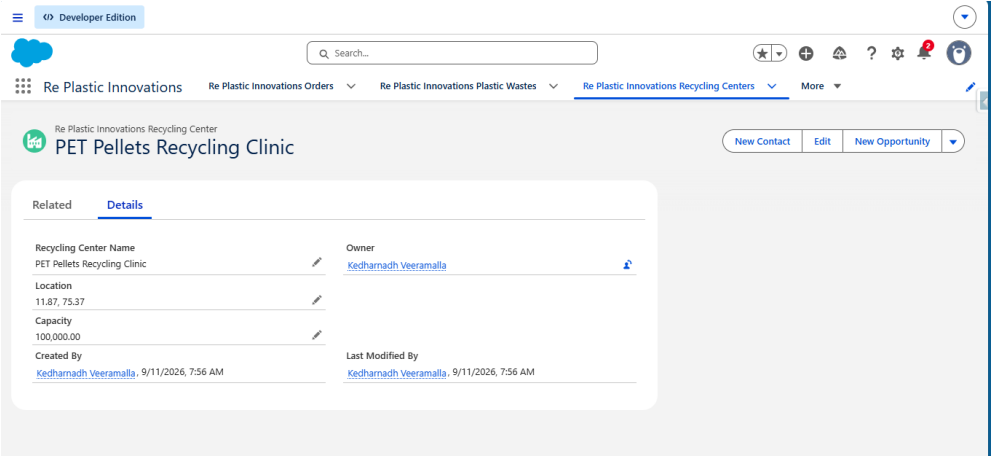


Fig. 8b – Recycling Center record



3.3 Usability

Usability was addressed through clear object names, direct navigation tabs, meaningful picklist values, validation messages, and automatic status calculations. The system provides immediate feedback when a user enters an invalid quantity or future collection date.

The stock-status formula also makes inventory conditions visible without requiring users to manually compare two fields. Automation reduces repetitive checking, while the role-based design limits users to the information relevant to their responsibilities.

PHASE 4 — DATA MIGRATION, TESTING & SECURITY

4.1 Data Handling

The project data model was created with dedicated fields for operational information such as weight, type, dates, quantities, stock levels, thresholds, locations, and statuses. Lookup relationships were used to maintain links between related records.

Data handling also includes automatic generation of record names where Auto Number fields were selected, which reduces manual naming effort and supports consistent record identification.

Figure 9: Recycled Product, Order and Restock Request Records

Fig. 9a – Recycled Product record

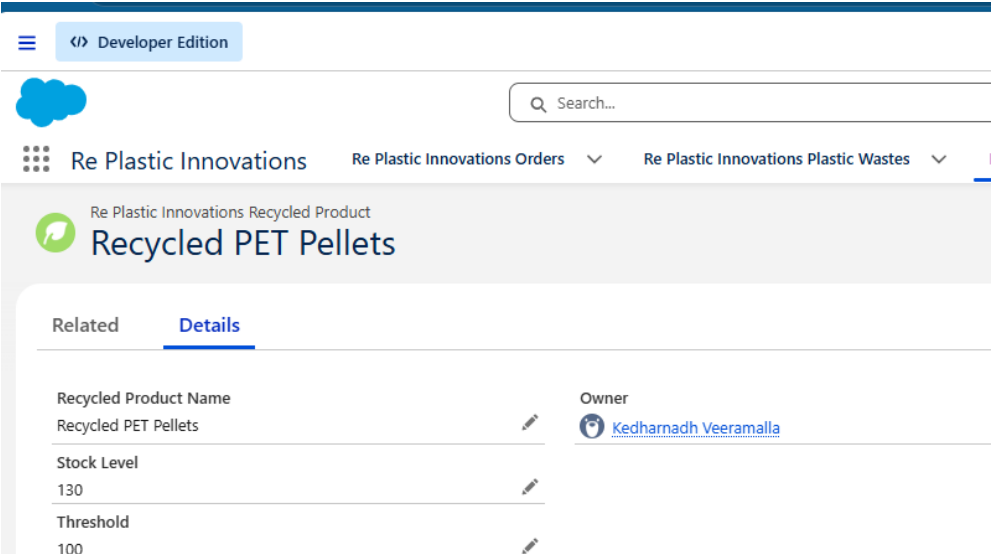


Fig. 9b – Customer Order record

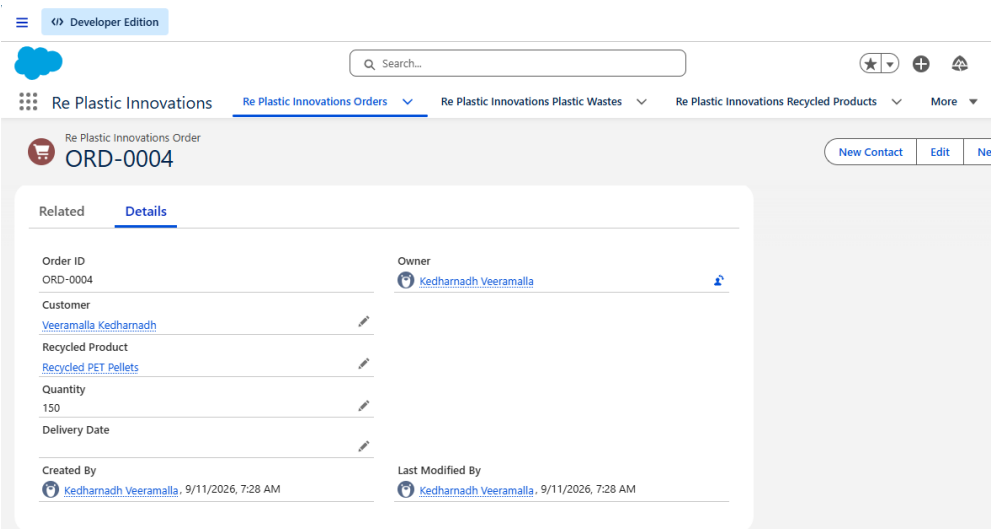
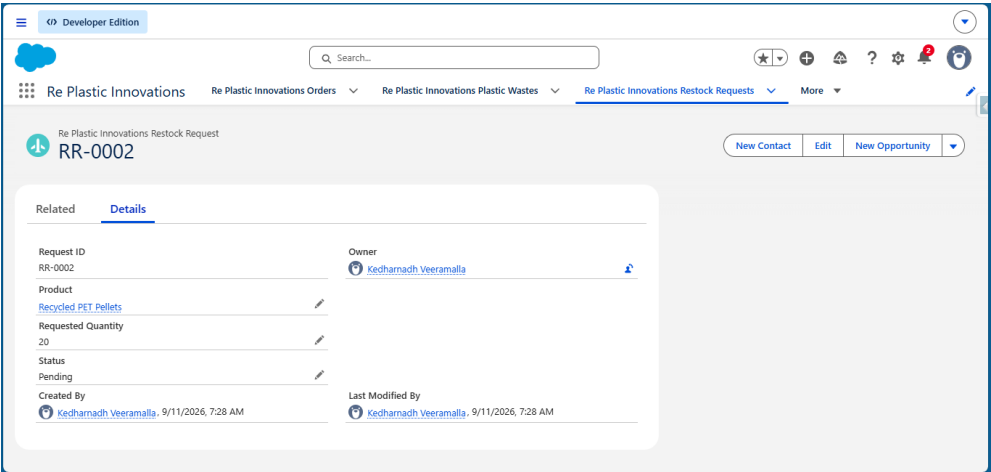


Fig. 9c – Restock Request record



4.2 Access Control

Access control was implemented at both object and record levels. Profiles define what users can do with objects, roles establish the organizational hierarchy, private organization-wide defaults restrict record visibility, and sharing rules provide the required additional access.

Control	Purpose	Implemented Example
Profiles	Object-level permissions	Platform 1, Platform 2, Platform 3
Roles	Hierarchy and sharing context	CEO → Recycling Manager / Sales Representative → Warehouse Supervisor
OWD	Default record visibility	Relevant custom objects configured as Private
Sharing Rules	Additional record access	CEO to Recycling Manager; CEO to Sales Representative; Sales Representative to Warehouse Supervisor

Figure 10: Record-Level Security and Sharing Configuration

Fig. 10a – Sharing Settings

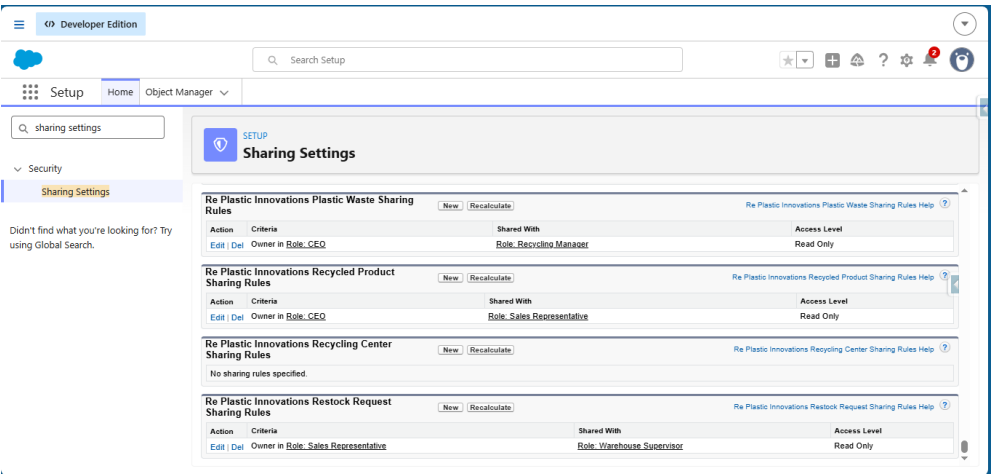


Fig. 10b – Sharing Rules

The screenshot shows the Salesforce Setup page for 'Developer Edition'. The left sidebar has 'Setup' selected, with 'Security' expanded and 'Sharing Settings' highlighted. The main content area is titled 'Sharing Settings' and displays a table of sharing rules:

Object	Sharing Method	Access Level	Checkmark
Web Cart Document	Private	Private	✓
Work Item	Private	Private	✓
Work Item Test Suite Group Execution	Private	Private	✓
Work Order	Private	Private	✓
Work Plan	Private	Private	✓
Work Plan Template	Private	Private	✓
Work Step Template	Private	Private	✓
Work Type	Private	Private	✓
Work Type Group	Public Read/Write	Private	✓
Re Plastic Innovations Order	Public Read/Write	Private	✓
Re Plastic Innovations Plastic Waste	Private	Private	✓
Re Plastic Innovations Recycled Product	Private	Private	✓
Re Plastic Innovations Recycling Center	Public Read/Write	Private	✓
Re Plastic Innovations Restock Request	Private	Private	✓

Other Settings: [Other Settings Help](#)

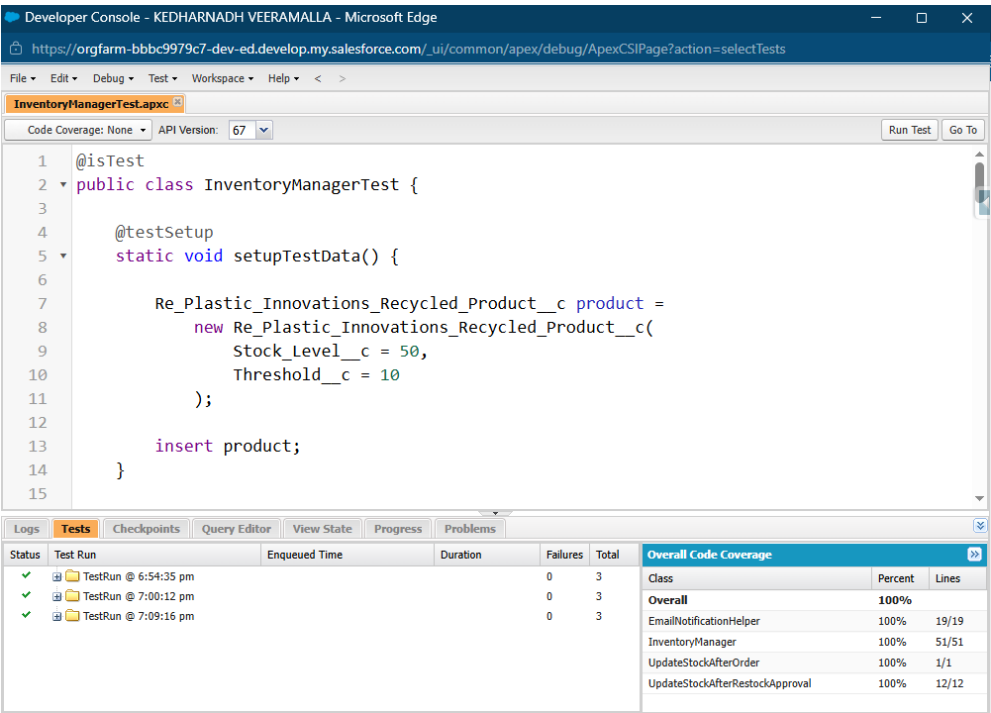
4.3 QA Testing

The Apex test class covers the core inventory scenarios required by the project. Test data starts with a recycled product having stock of 50 and threshold of 10. The tests then verify sufficient-stock order processing, insufficient-stock restock creation, and stock increase after restock approval.

Test Scenario	Expected Result	Observed Result
Order quantity within available stock	Stock decreases by ordered quantity	Passed
Order quantity exceeds available stock	Pending restock request is created for shortage	Passed
Restock request changes to Approved	Product stock increases by requested quantity	Passed

Figure 11: Apex Test Results and Code Coverage

Fig. 11a – Apex test results and code coverage



4.4 Data Integrity

Data integrity is supported by validation rules, formula logic, controlled picklist values, and Apex processing. Invalid order quantities are rejected, future collection dates are blocked, and stock changes are performed by the implemented business logic instead of relying on manual calculation.

The restock trigger also checks that the status has newly changed to Approved before processing, preventing the same approval event from being repeatedly processed by later edits.

Figure 12: Data-Integrity Validation

Fig. 12a – Plastic Waste validation rule

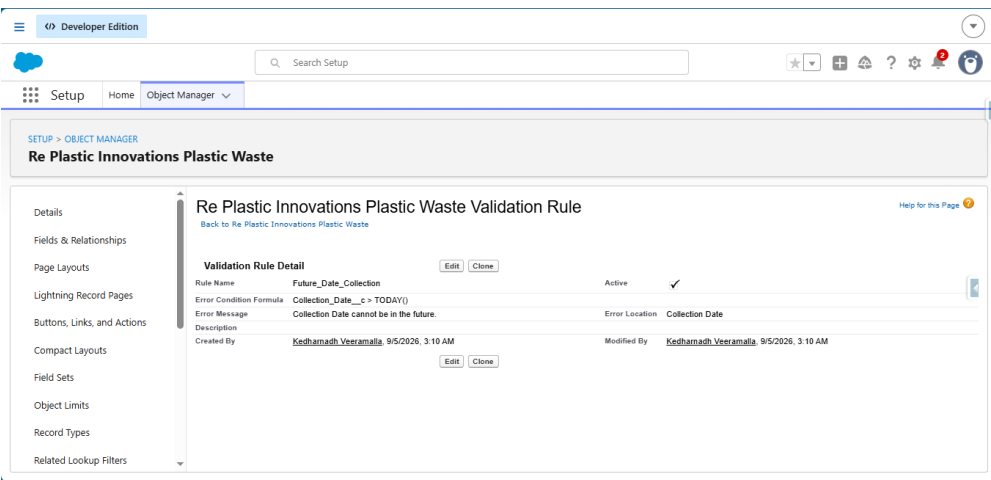


Fig. 12b – Order quantity validation

PHASE 5 — DEPLOYMENT, DOCUMENTATION & FINAL PRESENTATION

5.1 Troubleshooting

Troubleshooting focused on verifying field API names, lookup relationships, Flow references, trigger execution, stock calculations, email configuration, and test behavior. One important implementation detail was ensuring that the scheduled Flow assigned the generated Task to the product owner through the Owner ID.

Apex execution was verified using the project test scenarios. The order trigger and restock-approval trigger were kept focused on their respective business events so that each automation has a clear responsibility.

5.2 Adherence to Timelines

The work was organized progressively: configuration and data structure were completed first, followed by automation, Apex logic, security, and testing. This sequence allowed each later stage to use the components established in the earlier stages and reduced rework.

5.3 Innovation

The project combines Salesforce declarative capabilities with Apex processing to create a practical inventory workflow for recycled products. Instead of treating stock monitoring, shortage creation, restocking, and notification as separate manual activities, the implementation connects them into a single business process.

Figure 13: End-to-End RePlastix Inventory Workflow

Fig. 13a – Order processing

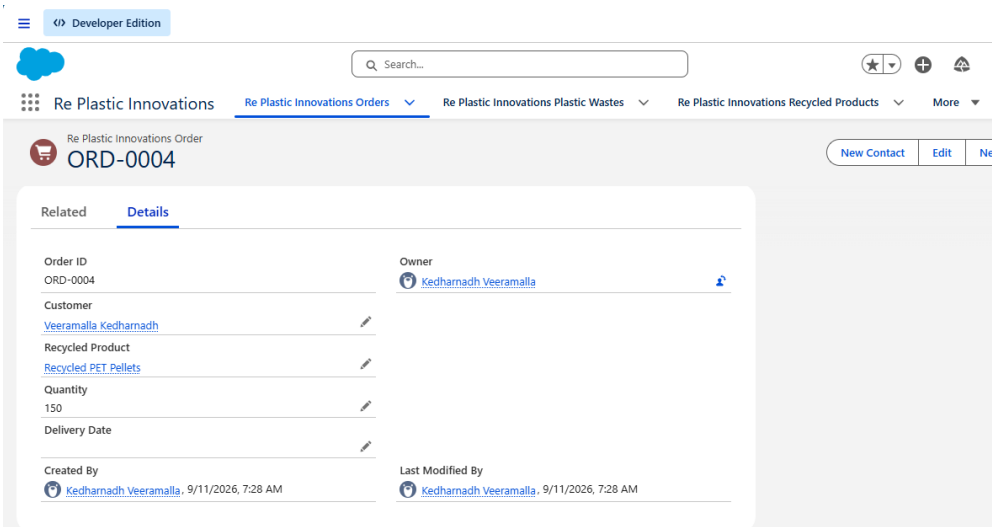


Fig. 13b – Restock approval

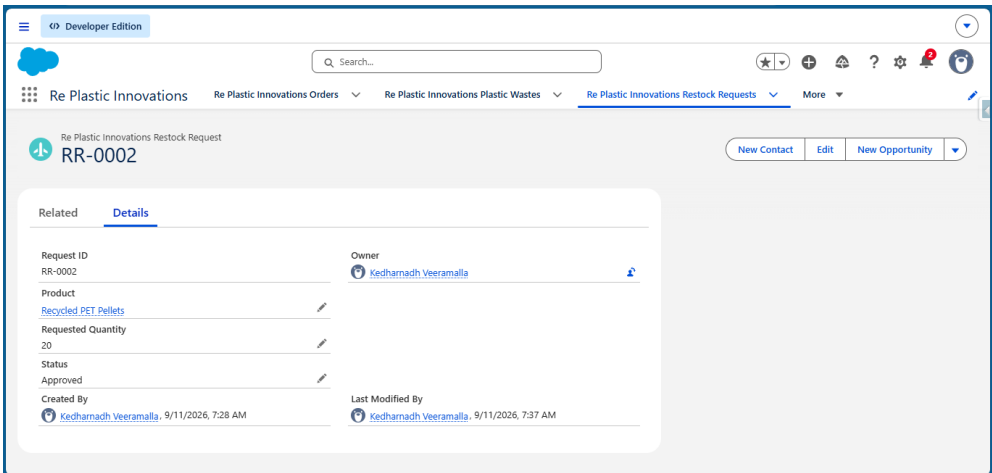
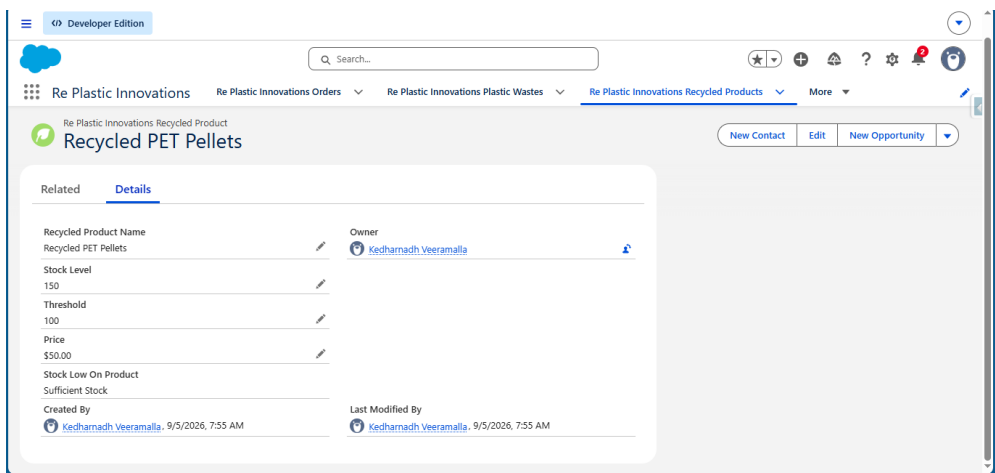


Fig. 13c – Updated product inventory



5.4 Document Functional Overview

Functionally, the system supports five connected areas: plastic-waste collection, recycling-center management, recycled-product inventory, customer orders, and restock requests. Users can maintain records through the Lightning application while automation handles repetitive monitoring and transaction processing.

The main operational scenario is: a customer order is created → available stock is checked → stock is reduced when sufficient → otherwise a Pending Restock Request is created → the request can be changed to Approved → product stock is increased and an email notification is sent.

5.5 Document Technical Blueprint

The technical design consists of custom Salesforce objects and lookup relationships at the data layer; a Lightning application and custom tabs at the interface layer; validation rules and a formula at the business-rule layer; Scheduled Flow for daily low-stock monitoring; and Apex classes/triggers for transaction-based inventory processing.

Layer	Implemented Components
Data	Plastic Waste, Recycling Center, Recycled Product, Order, Restock Request, Account lookup
UI	Re Plastic Innovations Lightning App and custom tabs
Declarative Logic	Stock status formula, two validation rules, scheduled low-stock Flow
Programmatic Logic	InventoryManager, two Apex triggers, EmailNotificationHelper
Security	Profiles, roles, private OWD, sharing rules
Testing	InventoryManagerTest covering three core scenarios

Figure 14: Apex Backend and Trigger Implementation

Fig. 14a – InventoryManager Apex class

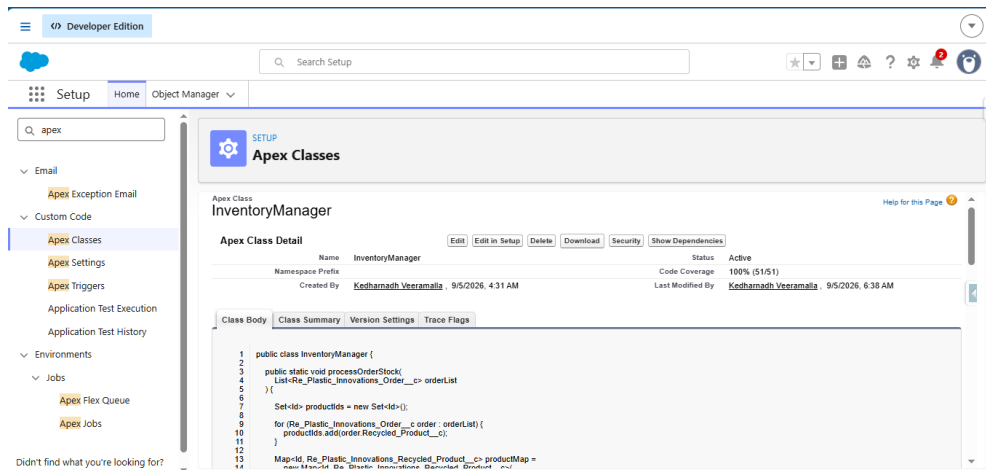
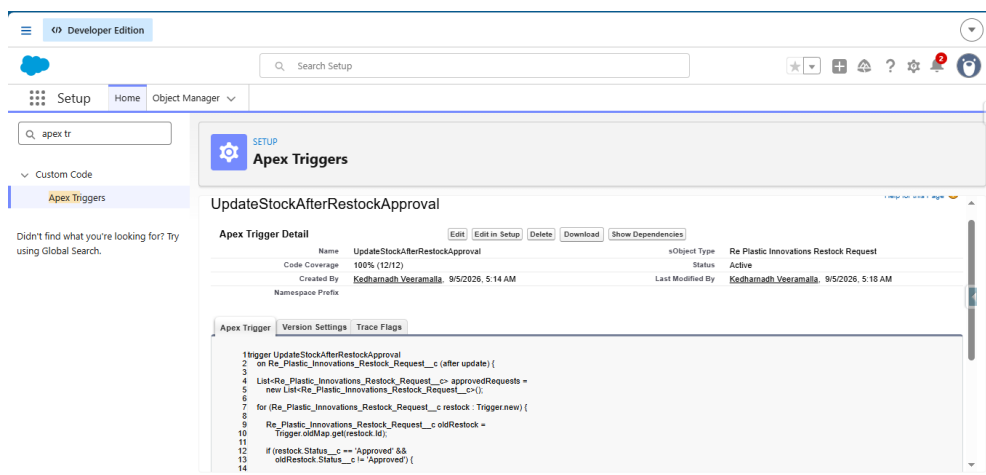


Fig. 14b – Restock approval trigger



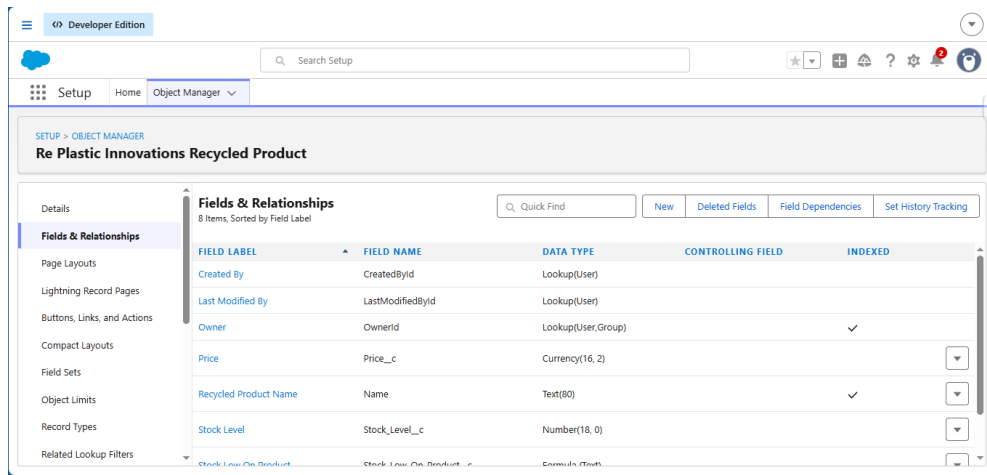
5.6 Document Setup Manual

The setup sequence for reproducing the implementation is:

- Create the five custom objects and required fields.
- Create lookup relationships and custom tabs.
- Create the Re Plastic Innovations Lightning application and add the required navigation items.
- Configure the stock-status formula and validation rules.
- Create the scheduled low-stock Flow and activate it.
- Create InventoryManager, the order trigger, the restock-approval trigger, and EmailNotificationHelper.
- Configure profiles, roles, private organization-wide defaults, and sharing rules.
- Create representative records and run the InventoryManagerTest class.

Figure 15: Salesforce Object Manager Configuration

Fig. 15a – Salesforce Object Manager – Recycled Product



5.7 Project Demo Video Planning

The demonstration video should be kept between two and five minutes and should focus on the working business scenario rather than showing every configuration screen.

- Introduce the student and project.
- State the problem and the Salesforce-based solution.
- Show the Re Plastic Innovations application and main navigation.
- Demonstrate a sufficient-stock order and show the stock reduction.
- Demonstrate an insufficient-stock order and show the automatically created Pending Restock Request.
- Change the request to Approved and show the stock increase and notification behavior.
- Briefly explain why Flow was used for scheduled low-stock monitoring and Apex triggers were used for transaction-based inventory processing.
- Finish by showing successful test results and summarizing the outcome.

5.8 Visual Demonstration

The final presentation should use screenshots that prove the important implemented features. The strongest evidence is a combination of application navigation, product inventory, automation, the shortage/restock workflow, security configuration, and successful testing.

Figure	
1	Lightning application and navigation
2	Role/profile or sharing configuration
3	Object/field configuration
4	Scheduled low-stock Flow
5	Order/restock trigger workflow
6	Formula and validation rules
7	Application record view
8	Navigation/relationships
9	Product/order/restock records
10	Security configuration
11	Test results and code coverage
12	Validation/data-integrity result
13	End-to-end workflow
14	Technical architecture
15	Setup configuration

5.9 Explanation Clarity

The project should be explained using a simple sequence: problem, solution, data model, automation, live scenario, technical choices, testing, and result. Technical terms should be connected to their purpose. For example, validation rules are used for simple data restrictions, Scheduled Flow is used for time-based low-stock checking, and Apex is used where the process requires custom logic across related records.

The restock process should be described accurately as a status-based workflow: a Pending request is created automatically, a user changes it to Approved, and the after-update trigger processes the approval. No Salesforce Approval Process should be claimed because that configuration was not implemented.

5.10 Realism & Quality

The solution represents a realistic operational scenario in which recycled-product demand affects available stock and shortages require replenishment. The combination of standard Salesforce capabilities, custom objects, controlled access, declarative automation, and Apex provides a practical implementation rather than a purely theoretical design.

Figure 16: Final RePlastix Application Records

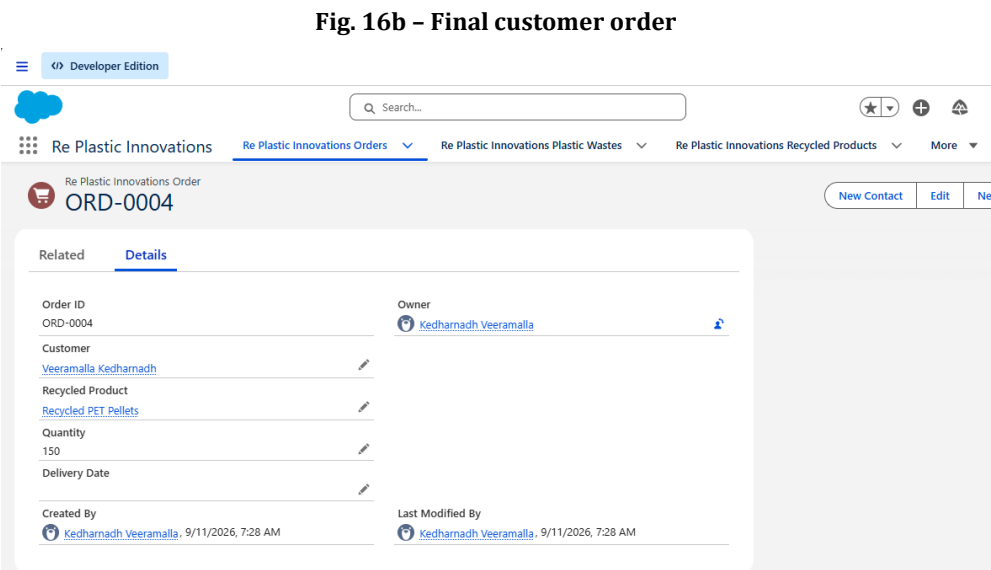
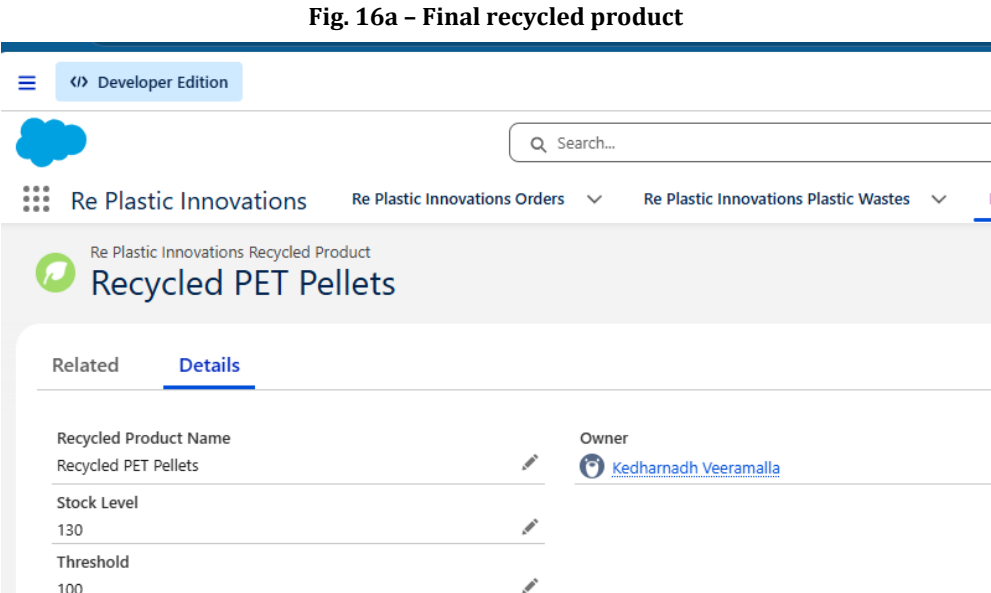
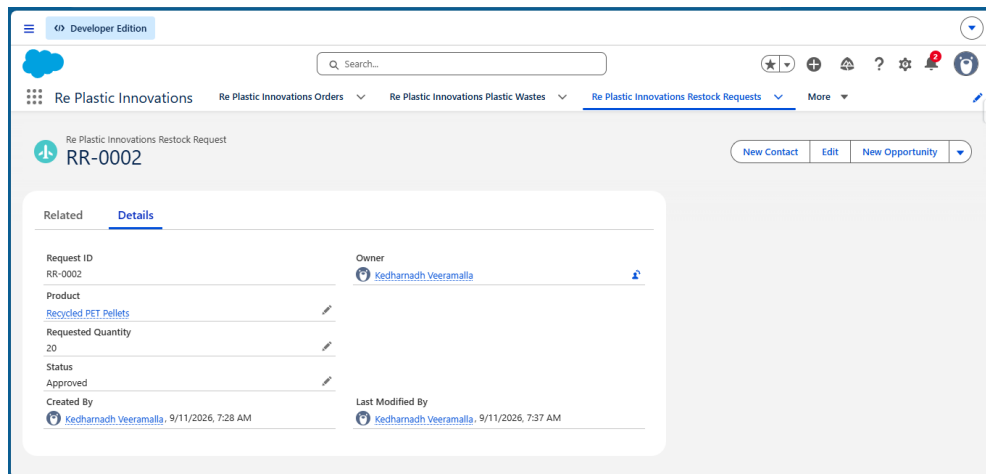


Fig. 16c – Final approved restock request



5.11 Scalability & Future Plan

The current architecture can be extended by adding dashboards and reports for management, more detailed recycling and processing stages, stronger notification preferences, richer inventory analytics, and additional automation as the volume of records grows. The separation of objects and the use of focused automation components provide a foundation for these future enhancements.

END OF PROJECT DOCUMENTATION

RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions